Drug Interaction Severity Prediction

Deep-Pharma Multimodal Intelligence DataHack 3.0

Kaggle GitHub

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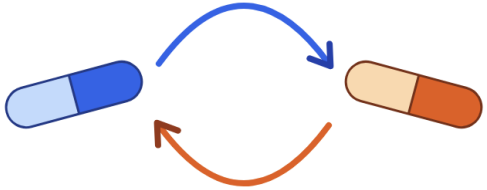
**Generative AI provided limited assistance in line with the course policy on usage of AI.*

Context & Terminology

- 1 in 4 U.S. adults filled 3+ prescriptions monthly (2015–2018)
- This practice polypharmacy, creates compounding Drug-Drug Interactions (DDI) risk
- Clinical trials cannot cover all possible drug combinations

- Drug-Drug Interaction (DDI) When one medication alters the effect of another, changing efficacy or causing adverse reactions
- Polypharmacy Concurrent use of 5 or more medications by a single patient



Research Questions

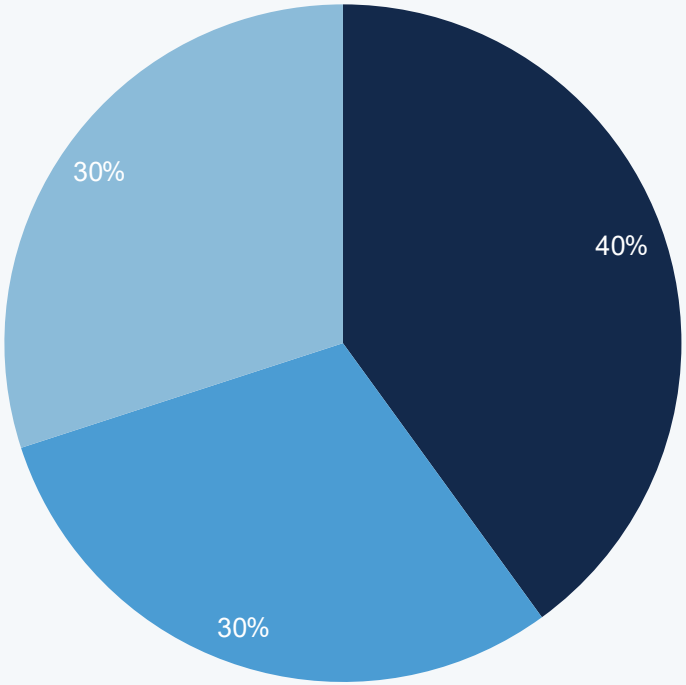
RQ1 — Supervised Machine Learning

- Can a multi-task ML model accurately predict DDI severity, adverse event occurrence, and PRR risk from pharmacological and molecular drug pair features?

RQ2 — Unsupervised Machine Learning

- What natural groupings exist within drug pair interaction profiles, and do these clusters align with meaningful differences in severity and side effect patterns?

Prediction Task Breakdown



■ Severity Classification (40%) ■ Side Effect Prediction (30%) ■ PRR Regression (30%)

Part 1: Severity Classification

Weight: 40% | Metric: Macro F1

3-class: Major / Moderate / Minor
Gradient Boosting — final model

Score

0.5666

Part 2: Side Effect Prediction

Weight: 30% | Metric: Micro F1

50-label binary prediction
MultiOutput Logistic Regression

Score

0.5772

Part 3: PRR Regression

Weight: 30% | Metric: Inv. RMSE

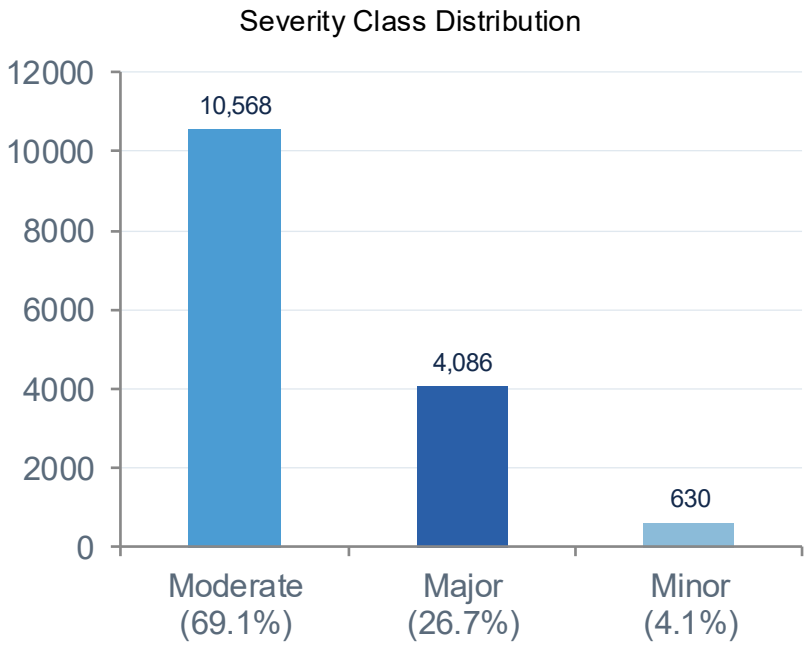
50-label continuous risk score
Ridge Regression (masked eval)

Score

0.2023

Exploratory Data Analysis

- Scale: Analyze the 15,284 training pairs versus 3,974 test pairs
- Target Distribution: Check for class imbalance in Severity (Major, Moderate, Minor).
- Label Sparsity: Evaluate the frequency of the 50 specific side effect labels.



Methodology

1

Raw Text Concat

Mechanism, pharmacodynamics, ADME, CYP450, indications, warnings, toxicity → single string per drug pair

2

Stratified Split

80/20 train/test split stratified on severity class to maintain class proportions

3

TF-IDF Vectorization

2,000 tokens, English stop word removal. Fit on training split only — no data leakage

4

Shared Feature Matrix

All three tasks use the same TF-IDF representation. SMILES included as raw text tokens

Models by Task

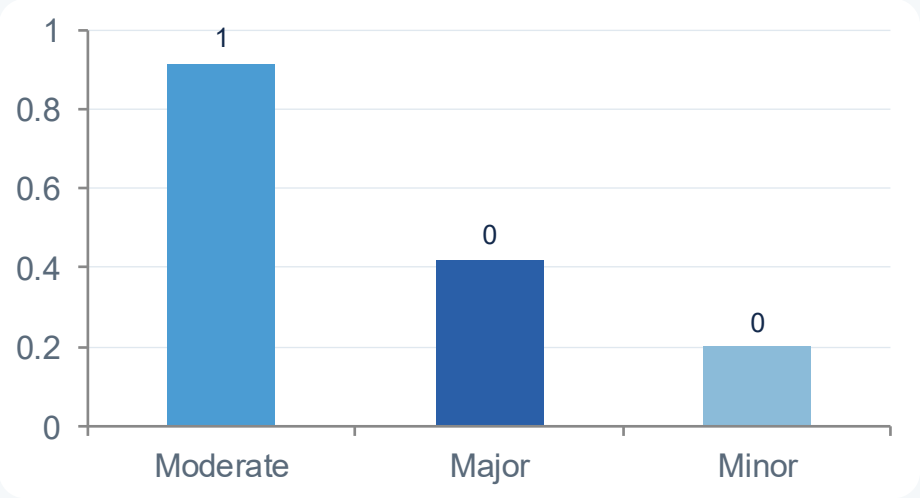
Severity Classification	Logistic Regression · Random Forest · Gradient Boosting (✓ selected)	*GB: n_estimators=100, max_depth=3, lr=0.1*
Side Effect Prediction (50 labels)	MultiOutputClassifier wrapping Logistic Regression — fit independently per label	*max_iter=1000*
PRR Regression (50 labels)	Multi-output Ridge Regression — predictions clipped at zero	*Alpha (default); sparse-stable*
Unsupervised Clustering	K-means on 50 binary AE columns — silhouette sweep k=2 to 6	*Random state fixed for reproducibility*

Supervised Severity Classification

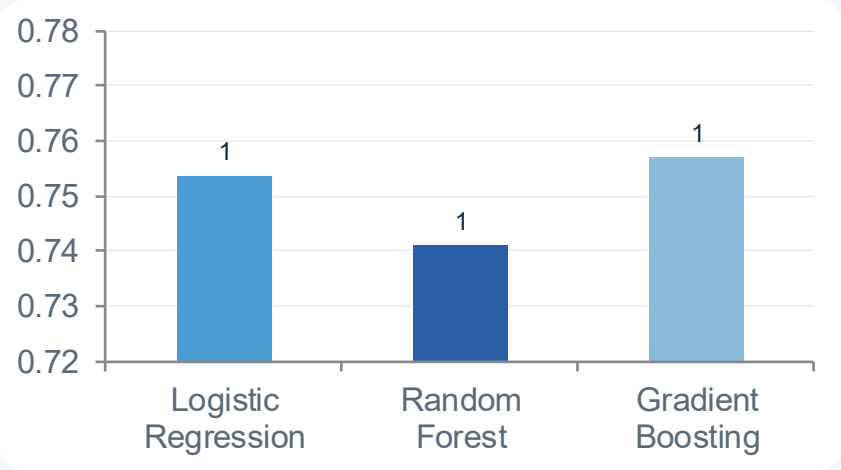
Model Comparison — Test Set

Model	Accuracy	Macro F1	AUC
Logistic Regression	0.7694	0.5395	0.9241
Random Forest	0.7661	0.5639	0.9146
Gradient Boosting ✓	0.7746	0.5666	0.9252

Per-Class Recall — Gradient Boosting



5-Fold Cross-Validation (Training Subset)

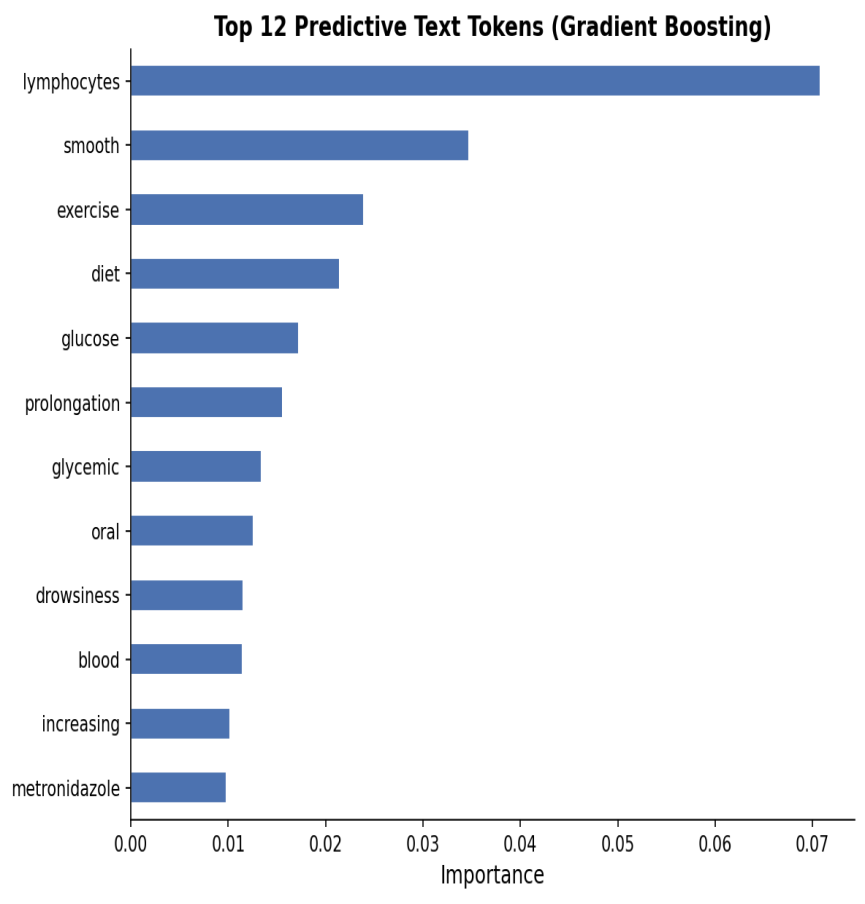


All models: AUC > 0.91. Strong class-level discrimination across unseen scaffolds.

AUC delta < 0.008 across all models — further gains require feature engineering, not model switching.

Minority class (Minor) recall only 19.8%. 16.8:1 imbalance drives this failure. Major recall 42–50%.

Feature Importance & Pharmacological Insights



Pharmacological Interpretation

"lymphocytes" (MDI: 0.07)

Immunosuppressants (e.g., tacrolimus, cyclosporine) heavily use this term — high DDI risk class

"glucose / glycemic / diet" (MDI: 0.017–0.022)

Diabetic medications alter glucose metabolism; DDIs here can cause dangerous hypoglycemia

"prolongation" (MDI: 0.016)

QT prolongation — cardiac arrhythmia risk flagged across many high-alert drug classes

"smooth" (MDI: 0.034)

Smooth muscle effects linked to calcium channel blockers and antispasmodics

"drowsiness / oral" (MDI: ~0.011)

CNS depressants + route of administration markers

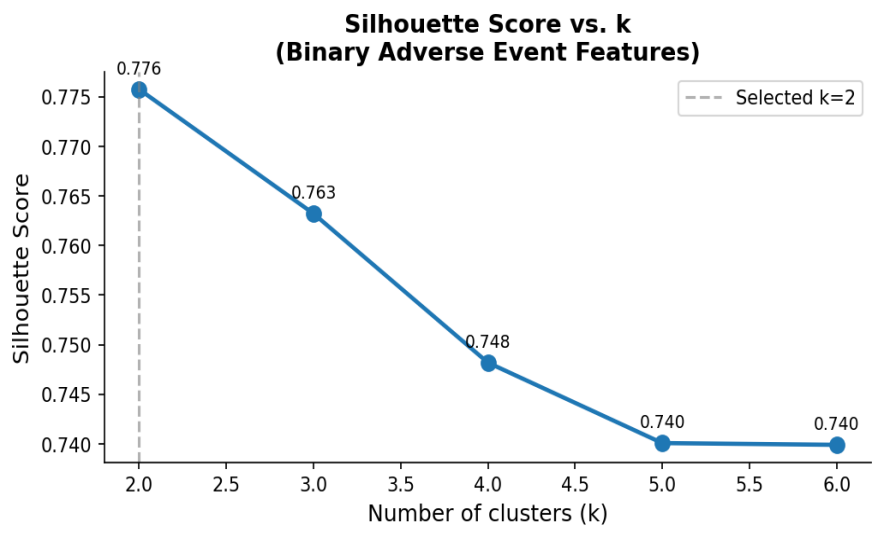
Additional Research: Smoking & Haemoglobin

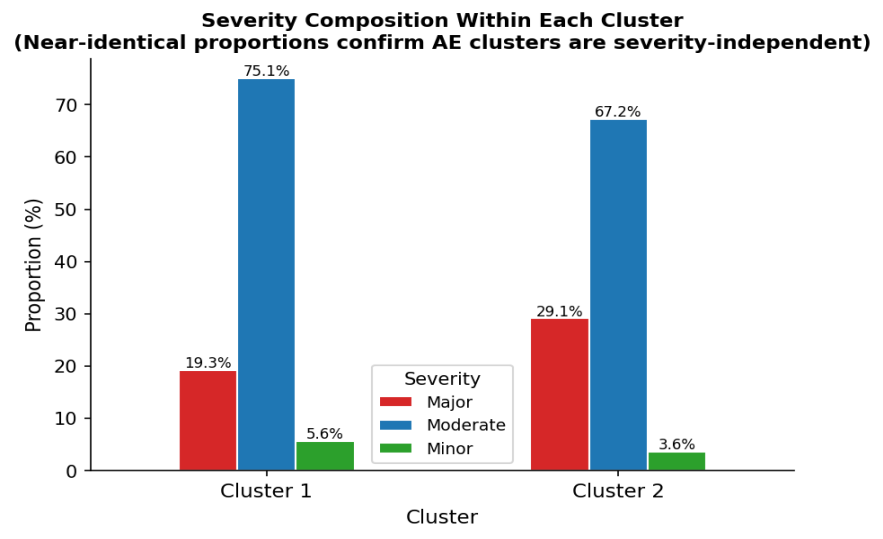
Haemoglobin decreased (AUC 0.890, F1 0.522) appeared among the 50 tracked AEs. Smoking is a known pharmacokinetic disruptor CYP1A2 induction from cigarette smoke accelerates metabolism of drugs like clozapine and theophylline, mimicking DDI effects.

Unsupervised Clustering

How Were the Clusters Determined?

K-means was applied to the 50 binary adverse event columns only (not severity or text features). K was selected by sweeping k=2 through k=6 and choosing the value with the highest silhouette score — a measure of how similar each point is to its own cluster vs. other clusters (higher = better). k=2 scored 0.776 and declined monotonically to 0.740 at k=6.





Key Finding (RQ2 — Meaningful Negative Result)

Cluster 1: high AE burden (prevalence 0.84–0.97). Cluster 2: near-zero (0.02–0.05). But severity composition is near-identical: Moderate = 75.1% vs 67.2%; Major = 19.3% vs 29.1%. Adverse event burden and interaction severity are independent dimensions.

Limitations & Future Directions

CYP450 Missingness

59.5% of Drug A and 47.1% of Drug B CYP450 profiles were missing. CYP overlap is a primary DDI predictor. Imputation or external CYP databases could recover this signal.

Label Independence

The MultiOutput classifier treats all 50 AE labels independently, ignoring co-occurrence structure (e.g., Nausea and Vomiting are highly correlated). Classifier chains could leverage this structure.

SMILES as Raw Text

SMILES strings encode molecular topology but TF-IDF tokenizes them as arbitrary characters. Molecular fingerprints (Morgan/ECFP) or graph neural networks would capture true structural similarity.

Cold-Start Test Set

The 3,974 held-out pairs have new molecular scaffolds not seen in training. Text-only models are structurally disadvantaged graph-based representations would generalize better to unseen chemistry.

CV Subset Limitation

5-fold CV was run on 3,000 observations for runtime efficiency. This may not fully represent the 12,227-observation training distribution.

Conclusions

01

Text features carry real clinical signal

TF-IDF on FDA FAERS pharmacological text achieved AUC > 0.91 for severity and AUC > 0.89 mean for adverse event prediction — strong performance on an unseen cold-start scaffold set.

02

Gradient Boosting outperformed all baselines

GB achieved the highest Macro F1 (0.5666) and AUC (0.9252). The narrow gap between models confirms the ceiling is features, not algorithms.

03

Class imbalance remains the dominant challenge

Minor class recall peaked at 19.8%. Addressing the 16.8:1 imbalance through SMOTE, class weighting, or threshold tuning is the most direct path to improvement.

04

AE clusters are severity-independent (RQ2)

K-means found high-burden vs. low-burden drug pair profiles, but severity composition was near-identical. Adverse event burden and severity are independent dimensions.

05

PRR requires structural features

Mean inverse RMSE of 0.2023 confirms TF-IDF is insufficient for continuous pharmacovigilance signal. Molecular fingerprints are the next step.

Anticipated Questions & Answers

Q: Why use Macro F1 instead of accuracy for severity?

A: Accuracy is misleading with a 16.8:1 imbalance — predicting "Moderate" for everything yields 69.1% accuracy with zero clinical value. Macro F1 weights all three classes equally, so the model is penalized for ignoring Minor and Major cases.

Q: Why didn't you use SMOTE or class weighting to fix the imbalance?

A: These are valid next steps listed in future work. The priority was establishing a clean baseline pipeline. Class weighting in GB is a direct and low-risk next step that would likely improve Minor recall substantially.

Q: How were the clusters determined?

A: K-means was applied to the 50 binary adverse event columns only. K was chosen by sweeping k=2 through k=6 and selecting the value with the highest silhouette score. k=2 scored 0.776 — higher cohesion and separation than any k from 3 to 6, which declined monotonically.

Q: Why was PRR so much weaker than the other tasks?

A: Three reasons: (1) only 21.5% of PRR entries are non-zero, making the signal sparse; (2) TF-IDF on SMILES lacks structural resolution for continuous regression; (3) high-variance events like Atrial Fibrillation have PRR values that are hard to predict from text alone.

Q: Why did the Kaggle score (0.5134) differ from your estimated composite (0.4605)?

A: The estimated composite was computed using my internal held-out test set. The Kaggle evaluation used a different scoring methodology and the actual competition test set. The divergence suggests my internal evaluation may have been conservative.

Q: What would be the highest-impact improvement?

A: Molecular fingerprints (e.g., Morgan/ECFP4 from RDKit). They encode true structural similarity and would improve both PRR regression and cold-start generalization to unseen scaffolds. Combined with CYP450 imputation from DrugBank, these two additions would likely lift all three task scores.